Vitae of Darren Shoemaker

Current as of September 2026

Academic History

Highest Degree Earned:	Ph.D., Forest Resources, 2025, Mississippi State University, Starkville, Mississippi
Additional Credentials:	M.S., Biology: Biology, 2021, Purdue University – Fort Wayne, Fort Wayne B.S., Biology: Ecology and Evolution, 2018, Indiana University – Purdue University, Fort Wayne Minor – Professional Writing, 2018, Indiana University – Purdue University, Fort Wayne

Work Experience/Appointments

Postdoctoral Research Associate. Department of Biological Sciences. University at Buffalo. Buffalo, NY. 2025-Present.

Graduate Research Assistant. Department of Wildlife, Fisheries and Aquaculture. Mississippi State University. Starkville, MS. 2021-2025.

Invasive Species Removal Specialist. Sanctuary Native Landscapes – Fort Wayne, Indiana. Summer 2021.

Graduate Teaching Assistant. Department of Biology. Purdue University – Fort Wayne. Fort Wayne, Indiana. 2019 – 2020.

Writing Consultant. Helmke Library. Purdue University – Fort Wayne, Fort Wayne, Indiana. 2015 – 2019.

Scholarly Activity

Manuscripts Published

1. Ahmad, H., **DJ Shoemaker**, B Liu, S Kolluru, E Mocek, P Dash. 2026. *Spatiotemporal water quality dynamics in Southern U.S. coastal reserves: A long-term perspective.* Regional Studies in Marine Science. DOI: https://doi.org/10.1016/j.rsma.2026.105445
2. Ong'ondo, F., H Ahmad, F Jose, SI Jhara, **DJ Shoemaker**, M Islam, P Dash, S Bhuyan, N Islam. 2026. *AI-Driven Geospatial Modeling of Urban Expansion and Environmental Impacts.* Remote Sensing in Earth System Sciences. DOI: https://doi.org/10.1007/s41976-026-00284-1
3. **Shoemaker, D.J.**, PC Smiley Jr., RB Gillespie. 2025. *Evaluating the Influence of Climatic and Hydrologic Variables on Fish Communities in Channelised Agricultural Headwater Streams in the Midwestern USA*. Freshwater Biology. DOI: https://doi.org/10.1111/fwb.70148
4. Ahmad, H., F Jose, P Dash, **DJ Shoemaker**, SI Jhara. 2025. *Hypoxia in the Gulf of Mexico: A machine learning approach for evaluation and prediction.* Regional Studies in Marine Science. DOI: https://doi.org/10.1016/j.rsma.2025.104363
5. Ahmad, H., F Jose, P Dash, **DJ Shoemaker**, SI Jhara. 2025. *Machine learning-based estimation of chlorophyll-a in the Mississippi Sound using Landsat and ocean optics data.* Environmental Earth Sciences. DOI: http://dx.doi.org/10.1007/s12665-025-12191-7
6. Miranda, L.E., R Krogman, **DJ Shoemaker**. 2024. *Assessment of pollution and impairments of US reservoirs based on expert opinion.* Frontiers in Environmental Science, section Freshwater Science. DOI: https://doi.org/10.3389/fenvs.2024.1488955
7. Ahmad, H, F Jose, **DJ Shoemaker**. 2024. *Mapping, Dynamics, and Future Change Analysis of Sundarbans delta using Cellular Automata and Artificial Neural Network Modeling.* IEEE Journal of Selected Topics in Applied Earth Observation and Remote Sensing. DOI: https://doi.org/10.1109/JSTARS.2024.3367116
8. Ahmad, H, M Abdallah, F Jose, HE Elzain, S Bhuyan, **DJ Shoemaker**, S Selvam. 2023. *Evaluation and mapping of predicted future land use changes using hybrid models in a coastal area.* Ecological Informatics. DOI: https://doi.org/10.1016/j.ecoinf.2023.102324

Software and Data Products

1. **Shoemaker, D.J.**, Miranda, S., and C. Dunn. Reservoir Impacts from Climate Change (ResClim). 2026. Version 1.0.0: U.S. Geological Survey software release. https://doi.org/10.5066/P13NURFZ

Conference Presentations

1. **Shoemaker, D.J.**, Smiley, P.C., Ruppert, K.M., and Gillespie, R.B. 2026. *Occupancy Patterns of Headwater Stream Fishes in an Agricultural Watershed.* Society for Freshwater Science Annual Meeting. Spokane, Washington.
2. Zachary Kellogg, Betsy Yankowiak, Bruce Kingsbury, **Darren Shoemaker**, Mark A. Jordan. 2025. *Spatial Analysis and Habitat Use of Snapping Turtles in an Urban Wetland Complex.* Mississippi Amphibian and Reptile Symposium. Jackson, MS.
3. **Shoemaker, D.J.** and Miranda, L.E. 2025. *Projected future reservoir impairment in the conterminous United States following three climate change scenarios.* Mississippi Chapter of the American Fisheries Society 51st Annual Meeting. Starkville, MS.
4. **Shoemaker, D.J.** and Miranda, L.E. 2024. *Predicted Impairment of Reservoirs in the United States under Three Climate Change Scenarios.* Graduate Student Symposium at the Dauphin Island Sea Lab. Dauphin Island Sea Lab, AL.
5. **Shoemaker, D.J.**, Ahmad, H., Miranda, L.E. 2024. *Modeling reservoir impairment from climate change using human dimensions.* Mississippi Chapter of the American Fisheries Society 50th Annual Meeting. Tupelo, MS.
6. **Shoemaker, D.J.**,Miranda, L.E., Dunn, C., Hunt, K.M., and Boyles, R.P. 2023. *Challenges Evaluating Reservoir Fish Habitats in a Changing Climate.* 153rd American Fisheries Society Annual Meeting. Grand Rapids, MI.
7. **Shoemaker, D.J.**, Miranda, L.E., Dunn, C., Hunt, K.M., and Boyles, R.P. 2023. *Vulnerability of Reservoir Fish Habitats to Climate Change.* Mississippi Chapter of the American Fisheries Society 49th Annual Meeting. Natchez, MS.
8. **Shoemaker, D.J.**, Gillespie, R, Smiley, P.C.2022. *Assessing the relationships of fish communities with climate and hydrology in agricultural headwater streams.* Mississippi Chapter of the American Fisheries Society 48th Annual Meeting. Hattiesburg, MS.
9. **Shoemaker, D.J**, and Curtis, G.L. 2019. *Influence of Agricultural Fields on Fish Assemblages within Channelized Streams.* Indiana Academy of Science 134th Annual Academy meeting. Indianapolis, IN.
10. **Shoemaker, D.J.** and Curtis, G.L. 2019. *Influence of Agricultural Fields on Fish Assemblages within Channelized Streams.* Society for Ecological Restoration Midwest Great Lakes Chapter Meeting. Pella, IA.

Grantsmanship: Pending $229,922

1. Project Sponsor: University at Buffalo (Internal)
   Program: College of Arts of Science AI Catalyst Grant Program
   Project Title: Sound of the streambed: AI-driven sonar assessments of freshwater habitats
   Investigators: C.A. Krabbenhoft (PI), **D. Shoemaker**
   Funding attepted: **$9,922**
   Timeline: Fiscal Year 2027
   Objectives: Catalyst grant to obtain sonar equipment to support ongoing research in the Lake Erie streams.
   Note: Co-wrote with PI as only faculty are eligible for internal awards

2. Project Sponsor: National Science Foundation
   Program: Postdoctoral Research Fellowship in Biology
   Project Title: Side scan sonar for instream fish habitat in the United States and Puerto Rico
   Investigators: **D.J. Shoemaker (PI)**
   Funding attempted: **$220,000**
   Timeline: 1/1/2028-12/31/2029
   Objectives: Implement side scan sonar surveys at selected National Ecological Observatory Network (NEON) sites to evaluate instream structure and substrate.

Grantsmanship: Not Funded

1. Project Sponsor: Association of Fish and Wildlife Agencies
   Program: Multistate Conservation Grant Program
   Project Title: Development of side-scan sonar surveys for sportfish habitat conservation
   Investigators: C.A. Krabbenhoft (PI), **D. Shoemaker**
   Funding attempted: **$242,705**
   Timeline: 1/1/2027-12/31/2027
   Objectives: Implement side scan sonar surveys with multistate partners to identify critical sportfish habitat.

Manuscript Reviews

1. International Journal of Biometeorology
2. Freshwater Biology
3. Environmental Monitoring and Assessment
4. Frontiers in Freshwater Science
5. Journal of Great Lakes Research

Teaching Activity

Course Instruction

Academic Year	Course Prefix and Number	Course Title	Role	Contact Hours per Week	Total Hours
Fall 2020	BIOL11170-Lab	Principles of Ecology & Evolution	TA	6	60
Summer II 2020	BIOL11170-Lab	Principles of Ecology & Evolution	TA	3	12
Spring 2020	BIOL11190-Lab	Principles of Structure and Function	TA	6	60
Fall 2019	BIOL11170-Lab	Principles of Ecology & Evolution	TA	6	60

Guest Lectures

1. *Real-world research in stream ecology.* 2026. Stream Ecology. University at Buffalo. Buffalo, New York.
2. *Osteological Correlates and Diet.* 2026. The Dinosaurs. University at Buffalo. Buffalo, New York.
3. *Data-driven ecology: conservation in the 21st century.* 2025. Ichthyology. University at Buffalo. Buffalo, New York.
4. *Arthropod identification in iNaturalist.* 2020. Purdue University - Fort Wayne, Fort Wayne, IN.

Workshops Led

2025 - Integrating GitHub and RStudio.

Other Teaching Activity

2020 - Developed and Implemented Genetic Barcoding Semester Project 2020 (BIOL11700, Principles of Ecology & Evolution)

2020 - Recorded online pre-lab lectures and lab exercise walkthroughs (BIOL11700, Principles of Ecology & Evolution)

2020 - Migrated lab materials to online format for virtual instruction during COVID-19 pandemic (BIOL11900, Principles of Structure and Function)

2019 - Updated lab manual and other materials for lab instruction (BIOL11700, Principles of Ecology & Evolution)

2019 - Led undergraduate student groups with Indiana Department of Natural Resources boat captains for river ecological surveys

Honors and Awards

2026

- University at Buffalo Art of Research competition finalist. Buffalo, New York.

2025

- Best Student Oral Presentation. American Fisheries Society Mississippi Chapter Meeting. Starkville, Mississippi.

2024

- David H. Nabi Memorial Award. Mississippi State University Department of Wildlife, Fisheries & Aquaculture. Starkville, Mississippi.
- Graduate Student Symposium at the Dauphin Island Sea Lab Oral Presentation Honorable Mention. Dauphin Island Sea Lab, Alabama.
- Graduate Student Association Spring Research Symposium Third Place Oral Presentation (Ph.D. level). Mississippi State University, Starkville, Mississippi.

2023

- Graduate Student Association Fall Research Symposium Third Place Oral Presentation (Ph.D. level). Mississippi State University, Starkville, Mississippi.
- Best Student Poster Presentation. American Fisheries Society Mississippi Chapter Meeting, Natchez, Mississippi.
- Graduate Student Association Spring Research Symposium Best Student Poster (Ph.D. level). Mississippi State University, Starkville, Mississippi.

2018

- Indiana University – Purdue University Top 50 Award. Indiana University – Purdue University Fort Wayne, Fort Wayne, Indiana.
- Beverly Hume Memorial Award – "The Effects of Climate Change on Interaction between *Ursus arctos* and *Ursus maritimus*." College of Arts and Sciences, Indiana University – Purdue University Fort Wayne, Fort Wayne, Indiana.

Professional Development

Specialized Training and Workshops Attended

2024 - Adult CPR & AED Certification. USGS Cooperative Research Units: Emergency Care & Safety Institute

2023 - How Cultural Competency Drives Teamwork

2023 - Global Warming I: The Science and Modeling of Climate Change Coursera Certificate

2022 - Diverse Perspectives: Words are Powerful: Diversity Terminology Simplified

2022 - Safe Zone Awareness Training

2022 - Safe Zone Ally Training "Building Allies and Supportive Environments"

2022 - Safe Zone LGBTQ+ 101

2022 - Safe Zone Gender 101 (Asynchronous Training)

2019 - PFW Office of Graduate Studies Building Your Inclusive Classroom

2018 - PFW Graduate Assistant Workshop

Seminars and Meetings Attended

2024 - Amphibian and Reptile Conservation Meeting. Mississippi Department of Wildlife, Fisheries, and Parks, Museum of Natural History. Jackson, Mississippi.

2023 - Mississippi State University College of Forest Resources Diversity Roundtable Discussion

2022 - Leadership Seminar: Diverse Perspectives

2022 - Leadership Seminar: How Cultural Competency Drives Teamwork

Service

Public Service and Outreach

2025

Chaired selection Committee for 2025 David H. Nabi Memorial Award in Mississippi State University Department of Wildlife, Fisheries & Aquaculture.

Served as mock state legislator for Wildlife, Fisheries & Aquaculture Administration and Communications course.

2024

Served as evaluator for Summer 2024 Undergraduate Research Symposium.

2023

Presented research and career opportunities in natural resources to underrepresented high school seniors through Students Paving a Road to Knowledge (S.P.A.R.K).

2022-2024

Volunteer semesterly for Mississippi State University Wildlife, Fisheries & Aquaculture Graduate Student Association Adopt-a-Highway cleanup.

2021

Worked community tables for Purdue Fort Wayne Environmental Resources Center Earth Day celebration.

2018

Authored, developed, and produced radio PSAs for Mad Anthony's Children's Hope House.

Service and Support for Scientific Organizations

2022-2023

Assisted Mississippi State University graduate student Joshua Stafford with placing an acoustic sensor array around Eagle Lake, Steele Bayou, and Muddy Bayou, capturing, and implanting acoustic transmitters in silver carp, and active tracking of silver carp. Vicksburg, Mississippi.

2020

Volunteered radio tracking snapping turtles at Eagle Marsh. Little River Wetlands Project. Fort Wayne, Indiana.

2019

Assisted Ball State graduate student Angela Fletcher with small mammal trapping. Fort Wayne, Indiana.

Professional Memberships

Society for Freshwater Science 2026-Present

American Fisheries Society. 2021-2025

Indiana Academy of Science. 2019-2021

Society for Ecological Restoration Great Lakes Chapter. 2019-2021

Beta Beta Beta Honor Society. 2017-2021

Networking

GitHub: https://github.com/thedungeonecologist

OrcID: https://orcid.org/0009-0002-3581-3944

Google Scholar: https://scholar.google.com/citations?user=nojq_iIAAAAJ&hl=en&oi=ao

ResearchGate: https://www.researchgate.net/profile/Darren-Shoemaker

References

Professional references available on request.